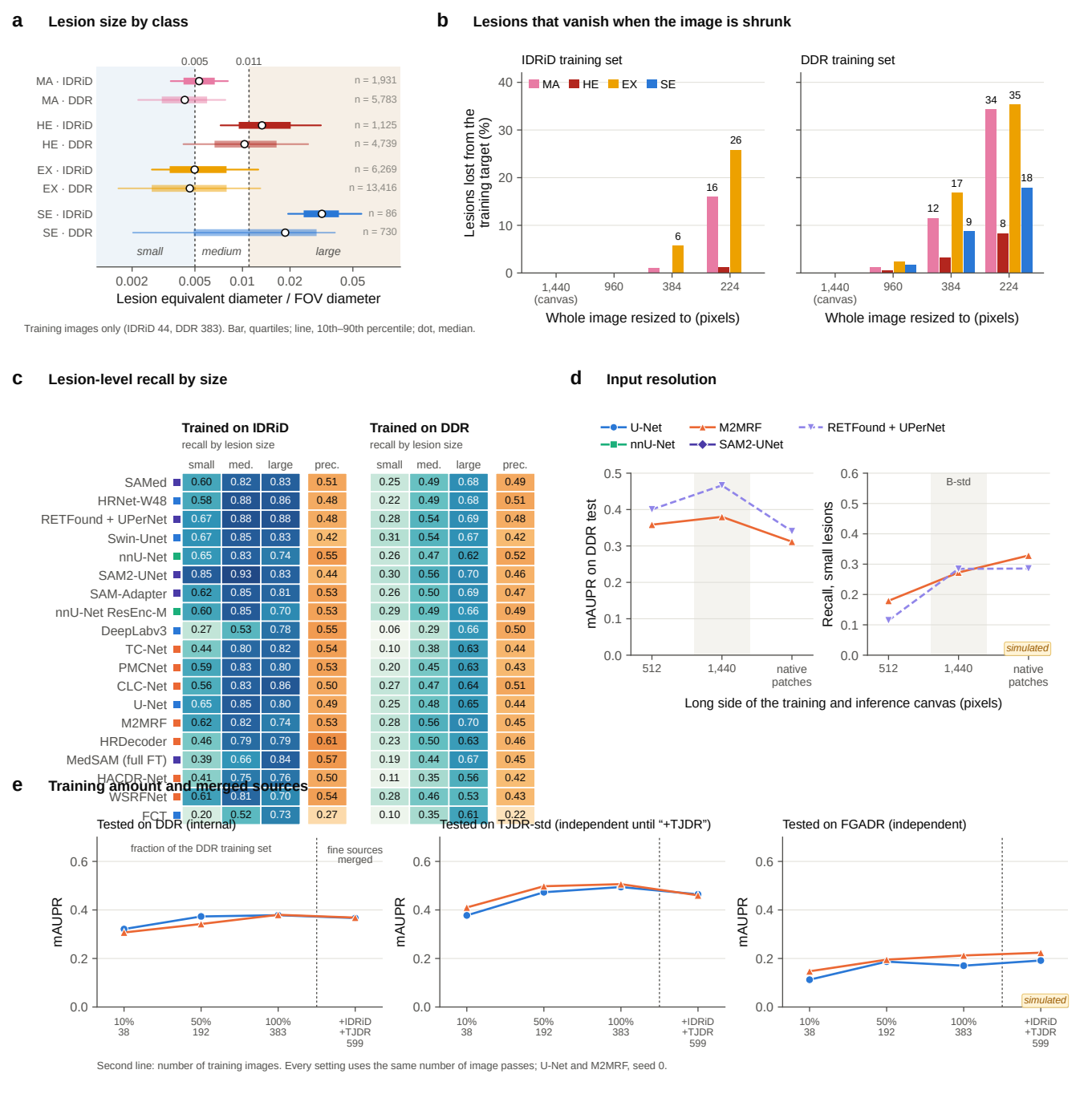
a
Lesion size by class
0.005
0.011
MA · IDRiD
MA · DDR
HE · IDRiD
HE · DDR
EX · IDRiD
EX · DDR
SE · IDRiD
SE · DDR
n = 1,931
n = 5,783
n = 1,125
n = 4,739
n = 6,269
n = 13,416
n = 86
n = 730
small
medium
large
0.002
0.005
0.01
0.02
0.05
Lesion equivalent diameter / FOV diameter
Training images only (IDRiD 44, DDR 383). Bar, quartiles; line, 10th–90th percentile; dot, median.
b
Lesions that vanish when the image is shrunk
IDRiD training set
DDR training set
MA
HE
EX
SE
Lesions lost from the training target (%)
1,440 (canvas)
960
384
224
Whole image resized to (pixels)
c
Lesion-level recall by size
Trained on IDRiD
recall by lesion size
Trained on DDR
recall by lesion size
small
med.
large
prec.
SAMed 0.60 0.82 0.83 0.51 | 0.25 0.49 0.68 0.49
HRNet-W48 0.58 0.88 0.86 0.48 | 0.22 0.49 0.68 0.51
RETFound + UPerNet 0.67 0.88 0.88 0.48 | 0.28 0.54 0.69 0.48
Swin-Unet 0.67 0.85 0.83 0.42 | 0.31 0.54 0.67 0.42
nnU-Net 0.65 0.83 0.74 0.55 | 0.26 0.47 0.62 0.52
SAM2-UNet 0.85 0.93 0.83 0.44 | 0.30 0.56 0.70 0.46
SAM-Adapter 0.62 0.85 0.81 0.53 | 0.26 0.50 0.69 0.47
nnU-Net ResEnc-M 0.60 0.85 0.70 0.53 | 0.29 0.49 0.66 0.49
DeepLabv3 0.27 0.53 0.78 0.55 | 0.06 0.29 0.66 0.50
TC-Net 0.44 0.80 0.82 0.54 | 0.10 0.38 0.63 0.44
PMCNet 0.59 0.83 0.80 0.53 | 0.20 0.45 0.63 0.43
CLC-Net 0.56 0.83 0.86 0.50 | 0.27 0.47 0.64 0.51
U-Net 0.65 0.85 0.80 0.49 | 0.25 0.48 0.65 0.44
M2MRF 0.62 0.82 0.74 0.53 | 0.28 0.56 0.70 0.45
HRDecoder 0.46 0.79 0.79 0.61 | 0.23 0.50 0.63 0.46
MedSAM (full FT) 0.39 0.66 0.84 0.57 | 0.19 0.44 0.67 0.45
HACDR-Net 0.41 0.75 0.76 0.50 | 0.11 0.35 0.56 0.42
WSRFNet 0.61 0.81 0.70 0.54 | 0.28 0.46 0.53 0.43
FCT 0.20 0.52 0.73 0.27 | 0.10 0.35 0.61 0.22
d
Input resolution
U-Net
M2MRF
RETFound + UPerNet
nnU-Net
SAM2-UNet
mAUPR on DDR test
Recall, small lesions
B-std
simulated
512
1,440
native patches
Long side of the training and inference canvas (pixels)
e
Training amount and merged sources
Tested on DDR (internal)
Tested on TJDR-std (independent until “+TJDR”)
Tested on FGADR (independent)
fraction of the DDR training set
fine sources merged
mAUPR
10% 38
50% 192
100% 383
+IDRiD +TJDR 599
simulated
Second line: number of training images. Every setting uses the same number of image passes; U-Net and M2MRF, seed 0.